

it's a proven technology in terms of field performance, quality, reliability and cost-to-performance ratio (best output power or capacity vis-a-vis the cost paid). The total cost-of-ownership (TCO) of VRLA batteries makes these the most preferred solution over DG sets and other options as return-to-investment recovery is fastest for VRLA batteries. The only limitation is that abusive conditions of use in India reduce the life of these batteries."

"Within VRLA, there are several technologies, and performance of the battery depends on many factors like design and construction of the battery, materials used for the positive and negative plates, grid and mesh, process and manufacturing technology, and valves," he adds.

"SMF VRLA batteries are convenient to use and transport compared to conventional flooded batteries. For UPS applications, it is the dominant battery type. In small solar LED lighting systems, lithium-ion batteries are replacing SMF VRLA, but their numbers at the moment are not large due to their high cost. We will see variants of VRLA battery like tubular gel and flat-plate gel batteries," informs Bruno.

Manufacturing in India. There are only a few big manufacturers of SMF VRLA batteries in India. At the same time, the domestic battery demand is expected to rise further. Apart from that, a lot of exports would take place due to environmental concerns in other countries. India is now exporting to many countries including the UK and the USA apart from the African continent.

Talking about manufacturing ecosystem, Bhatnagar shares, "India is very conducive to battery manufacture as the country has environmental systems and procedures in place and the manufacturers are putting a lot of thrust on ETP equipment."

Shah says, "We have seen the emergence of many VRLA battery manufacturers in the last five years. Even the existing manufacturers are further expanding their capacities. India has the best manufacturing environment due to the huge local demand. SMF

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VRLA batteries need to be charged and stored under recommended conditions, and it's a challenge to geographically transport the batteries across India. So it makes sense to not only produce these batteries but also have multiple warehousing/charging stations."

"The demand potential for SMF VRLA batteries is very large in India. Infrastructure growth rate makes the demand-supply gap even wider. The demand for electricity is increasing but the supply is limited. The most cost-effective solution is VRLA battery. That creates a huge market for VRLA. So even though there are several local makers, companies like Panasonic (which doesn't manufacture in India) will continue to do good business," he adds.

"There is a good scope for India to emerge as a VRLA battery manufacturing destination as manufacturing in China is becoming expensive due to currency appreciation and increased labour costs. Most of the raw materials are available indigenously and hence the dollar appreciation would be beneficial to manufacturers looking to export," adds Bruno.

Technology trends. Technology is improving with time. Batteries pass through many rough conditions like poor power quality while charging, deep discharge in some applications and use under 'partial state of charge' condition. Manufacturers are accordingly changing the recipe of the plates to match the new expectation or usage pattern.

"With regards to lead-acid batteries, manufacturers are working on variants using gel that will operate better in solar and UPS applications where high temperatures and deep

cycling are involved. We ourselves are working on flat-plate gel batteries that will be ready for the market by February next year," shares Bruno.

"Panasonic is focusing on longer-life batteries (especially for Indian conditions), special designs for batteries to be used for solar/windmills applications, and electric vehicle batteries for specific customers. We are looking at extending the average life of batteries from 3-3.5 years to five years or more," shares Shah.

Roadblocks. In India the main issue is availability of pure lead as there is only one producer of lead and most of the pure lead is imported. Another major component of SMF VRLA batteries is AGM separator. Unfortunately, there is not even a single AGM separator manufacturer in India, so it also is fully imported.

Bruno feels that the only roadblock for the battery industry is the overall economy. "I hope that the government does something fast to improve sentiments. As such, the industry is well positioned since demand is consistent and growing. However, for this demand to continue, it is important that the economy doesn't tank and consumer sentiments remain positive," he says.

"For Panasonic, the biggest challenge today is the service, logistics and charging infrastructure, especially because our strategic partners are located all over India. We are progressing and setting up the right system to clear all the bottlenecks. Also, in the near future we will set up testing, analysis, and pre- and post-installation support sites. Plus, we will have regular health check and monitoring systems for all our